value of $|\det(D\mathrm{T}_t)|$ can never be 0 (is is bounded from below by $\inf f/\sup g$). If everything is smooth, this means that the sign of the determinant is preserved and that no eigenvalue can ever be zero (this is also underlined in [87]). Unfortunately, the eigenvalues of $D\mathrm{T}_t$ are real at $t = 0$ (if we choose the Knothe map), are real at $t = \infty$ (since, as a gradient, the Jacobian is symmetric) but nothing prevents the eigenvalues to become complex and to pass from $]0,+\infty[$ to $]-\infty,0[$ without passing through 0 during the evolution! (even if this does not seem likely to happen, because the choice of $\mathbf{v}$ exactly forces T_t to become closer to the set of gradients, and gradients do not like complex eigenvalues).

A simple case, where there is no ambiguity if everything is smooth, is the 2D case. We can give a straightforward proposition.

Proposition 6.3. *Suppose that $(\xi_t, \mathbf{v}_t)$ is a smooth solution of the AHT system in the quadratic case, with $\mathrm{T}_t(x) = x - \xi_t(x)$ being a transport map between two given smooth densities $f, g \in \mathscr{P}(\Omega)$), where $\Omega \subset \mathbb{R}^2$ is a two-dimensional smooth convex and bounded domain. Suppose that $\det(D\mathrm{T}_0(x))$ is positive for every x. Also suppose that $\mathrm{T}_t \to \mathrm{T}_\infty$ in $C^1(\Omega)$ as $t \to \infty$ and that the evolution is non-trivial (i.e. there exists t such that $\mathrm{T}_t \neq \mathrm{T}_0$). Then T_∞ is the gradient of a convex function and is thus optimal.*

Proof. From the above considerations we have

$$M(T_0) - M(\mathrm{T}_t) = \int_0^t ||P[\xi_t]||^2_{L^2} \mathrm{d}t. \tag{6.1}$$

This implies that $\int_0^\infty ||P[\xi_t]||^2_{L^2} \mathrm{d}t < \infty$ and, at least on a sequence $t_n \to \infty$, we have $P[\xi_{t_n}] \to 0$ in $L^2(\Omega)$. Hence $P[\xi_\infty] = 0$ since $\xi_{t_n} \to \xi_\infty := x - \mathrm{T}_\infty(x)$ in C^1, and hence ξ_∞ and T_∞ are gradients. We need to prove that T_∞ is the gradient of a convex function, i.e. that the eigenvalues of $D\mathrm{T}_\infty$, which is now a symmetric matrix, are both positive (remember that we are in $\mathbb{R}^2$).

We know that $t \mapsto \det(D\mathrm{T}_t(x))$ is continuous in time, positive at $t = 0$, and cannot vanish. From C^1 convergence, this implies $\det(D\mathrm{T}_\infty(x)) > 0$ for every x. Hence, the eigenvalues of $D\mathrm{T}_\infty(x)$ have the same sign (for each fixed x).

Let us look now at $\mathrm{Tr}(D\mathrm{T}_\infty(x))$. This is a continuous function of x, and it cannot be zero. Indeed, if it vanished somewhere, at such a point the two eigenvalues would have different sign. Hence, the sign of this trace is constant in x. Note that this only works for $t = \infty$, as for $t \in]0,\infty[$ one could a priori have complex eigenvalues.

If we exclude the case where $\mathrm{Tr}(D\mathrm{T}_\infty(x)) < 0$ everywhere we have concluded. Yet, this case corresponds to T_∞ being the gradient of a concave function (remember again that in dimension two, positive determinant and negative trace imply negative-definite). And concave function are the worse possible transports for the quadratic case (see **Ex**(2)). This is impossible since $M(\mathrm{T}_\infty) < M(\mathrm{T}_0)$ (except if the curve $t \mapsto \mathrm{T}_t$ were constant), from (6.1). □ □

Note that the assumptions on the initial condition of the above theorem are satisfied both by the Knothe transport and by the Dacorogna-Moser transport map.

Unfortunately, the above proof is really two-dimensional as we characterized positive symmetric matrices as those which have positive trace and positive determinant.

Also, in practice it is almost impossible to guarantee the assumptions of the Theorem above and convergence can only be observed empirically.

Open Problem (convergence of AHT): justify, even assuming smoothness and very strong convergence, that the AHT flow converges to the optimal map, if initialized at Knothe and/or Dacorogna-Moser, in dimension higher than two, and prove a rigorous result with minimal assumption in 2D.

6.3 Numerical solution of Monge-Ampère

As we saw in last section, we can use for numerical purposes the fact that solving the quadratic optimal transport problem is equivalent to finding a transport map which is the gradient of a convex function. In the last section, an iterated method was produced where a sequence of transport maps T_n (it was a continuous flow, but in practice it is discretized), with the same image measure, were considered, and they were forced to converge to a gradient. Here the approach is somehow opposite: a sequence of gradient maps is considered, and the corresponding measures are forced to converge to the desired target.

More precisely, the idea contained in [218] is the following: consider the simplified case where the domain $\Omega = \mathbb{T}^d$ is a d-dimensional torus, the source measure ρ is a $C^{0,\alpha}$ density bounded from below and from above, and the target measure is the uniform measure with constant density 1; we look for a smooth function φ such that $D^2\varphi \leq \mathrm{I}$ (which is the condition for φ to be c-concave, with $c(x,y) = |[x-y]|^2$ on the torus, see Section 1.3.2) and

$$\det(\mathrm{I} - D^2\varphi) = \rho.$$

We stress the fact that, in this periodic setting, it is not possible to look for a convex function u satisfying $\det(D^2u) = \rho$. Actually, periodic convex functions do not exist (except constant ones)! The only possibility is to stick to the framework of c-concave functions, without passing to the convexity of $x \mapsto \frac{1}{2}|x|^2 - \varphi(x)$.

If we find such a function φ, then $\mathrm{T}(x) = x - \nabla\varphi(x)$ (in the periodic sense) will be the optimal map between ρ and $\mathscr{L}^d$. Note that the choice of a uniform target measure gives a non-negligible simplification: the correct equation with a generic target measure would involve a function of $x - \nabla\varphi(x)$ at the denominator in the right-hand side. This would make the following analysis much more involved.

The idea of Loeper and Rapetti in [218] is to solve the equation through an iterative scheme, based on a Newton method.

Box 6.4. **– Memo** – Newton's method to solve $F(x) = 0$

Suppose we seek solutions of the equation $F = 0$, where $F : X \to X$ is a smooth function from a vector space X into itself. Suppose a solution $\bar{x}$ exists, with $DF(\bar{x})$ which is an invertible linear operator. Given an initial point x_0, we can define a recursive sequence in this way: x_{k+1} is the only solution of the linearized equation $F(x_k) + DF(x_k)(x - x_k) = 0$. In other words, $x_{k+1} = x_k - DF(x_k)^{-1} \cdot F(x_k)$. This amounts at iterating the map $x \mapsto G(x) := x - DF(x)^{-1} \cdot F(x)$. This map is well-defined, and it is a contraction in a neighborhood of

$\bar{x}$. The iterations converge to $\bar{x}$ if the initial point belongs to such a suitable neighborhood. The order of convergence is quadratic, i.e. $|x_{k+1} - \bar{x}| \le C|x_k - \bar{x}|^2$. This is a consequence of the fact that the Lipschitz constant of G in a ball $B(\bar{x}, R)$ is proportional to $\max\{|F(x)| : x \in B(\bar{x}, R)\}$ and hence decreases as $O(R)$. To see this fact, just compute

$$DG = \mathrm{I} - DF^{-1} \cdot DF + DF^{-1} \cdot D^2F \cdot DF^{-1} \cdot F = DF^{-1} \cdot D^2F \cdot DF^{-1} \cdot F.$$

An alternative, and less "aggressive" method can make use of a step parameter $\tau > 0$, defining $x_{k+1} = x_k - \tau DF(x_k)^{-1} \cdot F(x_k)$. In this case the map $G(x) = x - \tau DF(x)^{-1} \cdot F(x)$ is a contraction but its contraction factor is of the order of $1 - \tau$, which only gives slower convergence (but could converge under weaker assumptions).

Here, if we have a guess φ and look for a solution $\varphi + \psi$, the linearization of the equation reads

$$\det(\mathrm{I} - D^2\varphi) - \mathrm{Tr}[\mathrm{cof}(\mathrm{I} - D^2\varphi)D^2\psi] = \rho,$$

where $\mathrm{cof}(A)$ denotes the cofactors matrix of a given square matrix A, satisfying $\mathrm{cof}(A) \cdot A = \det(A)I$.

Box 6.5. **– Memo –** Linearization of the determinant

It is well-known that $\det(\mathrm{I} + \varepsilon B) = 1 + \varepsilon \mathrm{Tr}[B] + O(\varepsilon^2)$ (this can be easily checked by expanding the determinant of $\mathrm{I} + \varepsilon B$ and looking at the terms which are first-order in ε). This gives the linearization of the determinant in a neighborhood of the identity matrix. If we look for the linearization of the determinant around another matrix A, and we suppose A invertible, we have

$$\begin{aligned}\det(A + \varepsilon B) = \det(A)\det(\mathrm{I} + \varepsilon A^{-1}B) &= \det(A)(1 + \varepsilon \mathrm{Tr}[A^{-1}B] + O(\varepsilon^2)) \\ &= \det(A) + \varepsilon \mathrm{Tr}[\mathrm{cof}(A)B] + O(\varepsilon^2).\end{aligned}$$

By continuity, the determinant and the map $A \mapsto \mathrm{cof}(A)$ being C^∞ functions, this last expression (avoiding the use of A^{-1}) is also valid for A non-invertible.

The iterative steps of the algorithm proposed in [218] are thus:

- Start with an arbitrary smooth function φ_0 such that $\mathrm{I} - D^2\varphi_0 > 0$ and fix a small value of $\tau > 0$. Typically, one takes $\varphi_0 = 0$.
- At every step n, compute $\rho_n := \det(\mathrm{I} - D^2\varphi_n)$.
- Define the matrix $M_n(x) := \mathrm{cof}(\mathrm{I} - D^2\varphi_n)$. Suppose that we can guarantee that M_n is positive-definite.

- Solve $\mathrm{Tr}[M_n D^2\psi] = \rho_n - \rho$, which is a second-order linear elliptic equation in φ, on the torus (i.e. with periodic boundary conditions). The solution is unique up to an additive constant (choose for instance the zero-mean solution) provided $M_n > 0$. Call ψ_n the solution.
- Define $\varphi_{n+1} = \varphi_n + \tau\psi_n$ and start again.

From the implementation point of view, the elliptic equation defining ψ, i.e. $\mathrm{Tr}[M_n D^2\psi] = \rho_n - \rho$ is solved in [218] via a finite-difference discretization of the second order derivatives. It is worthwhile mentioning that this kind of equation, i.e. $\mathrm{Tr}[MD^2\psi] = f$, where M is the cofactor matrix of an Hessian, can also be expressed in divergence form. Indeed, they read $\sum_{i,j} M^{ij}\psi_{ij} = f$, but $\sum_{i,j} M^{ij}\psi_{ij} = \sum_i \left(M^{ij}\psi_j\right)_i$ since $\sum_i M_i^{ij} = 0$ for all j (see **Ex**(43)). This allows to give a variational structure to this elliptic equation and solve it via finite-element methods. On the other hand, the condition $\rho_n := \det(\mathrm{I} - D^2\varphi_n)$ is more naturally interpreted in terms of second-order finite differences. Let us stress that one of the main difficulties of these equations is exactly the discretization of a non-linear operator such as the Monge-Ampère operator (i.e. the determinat of the Hessian).

The convergence of the algorithm is guaranteed by the following theorem, under regularity assumptions

Theorem 6.4. *Suppose $\rho \in C^{0,\alpha}(\mathbb{T}^d)$ is a strictly positive density and that τ is smaller than a constant c_0 depending on $||\log\rho||_{L^\infty}$ and $||\rho||_{C^{0,\alpha}}$. Then the above algorithm provides a sequence φ_n converging in C^2 to the solution φ of* $\det(\mathrm{I} - D^2\varphi) = \rho$

Proof. The proof is based on a priori bounds derived from both elliptic regularity theory (see [185], for instance) and Caffarelli's regularity for Monge-Ampère (see Section 1.7.6; for the periodic boundary conditions, see also [128]).

Choosing τ small enough and suitable constants $c_1 > 0$ and C_1, C_2, C_3 large enough, one can prove by induction the following statement: at every step n, we have $\mathrm{I} - D^2\varphi_n > c_1\mathrm{I}$, $||\log(\rho_n/\rho)||_{L^\infty} \le C_1$, $||\rho_n - \rho||_{C^{0,\alpha}} \le C_2$ and $||\varphi_n||_{C^{2,\alpha}} \le C_3$. To do that, note that we have $||\psi_n||_{C^{2,\alpha}} \le C||\rho - \rho_n||_{C^{0,\alpha}}$ from standard elliptic regularity theory (see Schauder Estimates in Chapter 6 in [185]). We can compute

$$\begin{aligned} \rho_{n+1} = \det(\mathrm{I} - D^2\varphi_n - \tau D^2\psi_n) &= \rho_n - \tau\mathrm{Tr}[\mathrm{cof}(\mathrm{I} - D^2\varphi_n)D^2\psi_n] + r_n \\ &= (1-\tau)\rho_n + \tau\rho + r_n, \end{aligned}$$

where r_n is the second-order rest in the linearization of the determinant, i.e. it involves second derivatives of φ_n and second derivatives of ψ_n, but is of order 2 in $\tau D^2\psi_n$. Hence, from the algebra properties of the $C^{0,\alpha}$ norm, we have $||r_n||_{C^{0,\alpha}} \le C\tau^2||D^2\psi_n||^2_{C^{0,\alpha}} \le C\tau^2||\rho_n - \rho||^2_{C^{0,\alpha}}$. Then, from $\rho_{n+1} - \rho = (1-\tau)(\rho_n - \rho) + r_n$ we get

$$||\rho_{n+1} - \rho||_{C^{0,\alpha}} \le (1-\tau)||\rho_n - \rho||_{C^{0,\alpha}} + C\tau^2||\rho_n - \rho||^2_{C^{0,\alpha}} \le ||\rho_n - \rho||_{C^{0,\alpha}}(1 - \tau + C\tau^2).$$

It is enough to chose τ small enough so that $1 - \tau + C\tau^2 < 1$, to go on with the bound on $||\rho_{n+1} - \rho||_{C^{0,\alpha}}$. For the L^∞ bound, we use again

$$\rho_{n+1} = (1-\tau)\rho_n + \tau\rho + r_n \le ((1-\tau)e^{C_1} + \tau + C\tau^2)\rho.$$

Again, for τ small enough and $C_1 > 0$, one can guarantee $(1-\tau)e^{C_1} + \tau + C\tau^2 \leq e^{C_1}$. The lower bound works in the same way.

In order to study the bound on φ_{n+1}, we write $\varphi_{n+1} = \varphi_n + \tau\psi_n$. Note that, from $\mathrm{I} - D^2\varphi_n \geq c_1\mathrm{I}$ and $|D^2\psi_n| \leq C$, a suitable choice of τ allows to guarantee the convexity condition $\mathrm{I} - D^2\varphi_{n+1} > 0$. Hence, the uniform bound $||\varphi_{n+1}||_{C^{2,\alpha}} \leq C_3$ simply follows from Caffarelli's theory, and from the bounds on ρ_{n+1}. And, finally, an upper bound on the Hessian of φ_{n+1} also implies a lower bound on $\mathrm{I} - D^2\varphi_{n+1}$ because of the determinant condition.

This shows that the bounds are preserved in the iterations, and the convergence comes from $||\psi_n||_{C^{2,\alpha}} \leq C||\rho - \rho_n||_{C^{0,\alpha}}$, where the norm $||\rho - \rho_n||_{C^{0,\alpha}}$ goes exponentially to 0. $\square$ $\square$

We finish this section by mentioning that, after this first work by Loper and Rapetti, which indeed provided interesting numerical results, many improvement have occurred. First, we underline that [280] extended to the case of a non-constant target density (but still, Lipschitz regularity of the target density was needed). Then, we remark that another strong limitation of [218] is given by its periodic boundary conditions. When working on different domains than $\mathbb{T}^d$ the situation is much trickier. It has been successfully studied by Benamou, Froese and Oberman in [42, 43], by using a monotone discretization of the Monge-Ampère operator. Monotonocity allows to prove the existence of a solution of the discretized problem, and to prove convergence towards the viscosity solution of the limit Monge-Ampère equation. Finally, [42, 43], also attacked the problem of the boundary conditions, and of the shape of the domain. With the approach by viscosity solutions, the authors are able to consider non convex and even non-connected source domains (hence allowing vanishing densities). They handle the second boundary value problem writing the target domain as the level set of a convex function. Yet, we prefer not to enter into details of these recent results, because of their technical difficulty and of the notions about viscosity solutions that we did not develop here.

6.4 Discussion

6.4.1 Discrete numerical methods

We strongly used in Chapter 1 the linear and convex structure of the Kantorovich problem, which is indeed a linear programming problem in infinite dimension. It is natural to look for methods which exploit this fact, after discretization. Here discretization simply means replacing μ and ν with two finitely atomic measures $\mu = \sum_{i=1}^N a_i\delta_{x_i}$ and $\nu = \sum_{j=1}^M b_j\delta_{y_j}$. For fixed N and M, the choice of "optimal" points x_i and y_j and of weights a_i, b_j so that the approximation is as precise as possible is a delicate matter that we do not want to discuss here. It is linked to quantization problems for the measures μ and ν (see [188] for a survey of the theory from a signal processing point of view, and [74] for a variational asymptotical approach). We note anyway that the difficult issue is the choice of the points, since once the points are fixed, the natural choice for the weights is $a_i = \mu(V_i)$, $b_j = \nu(W_j)$, where V and W represent the Voronoi cells of the points $(x_i)_i$ and $(y_j)_j$, respectively (see below).